Remote Voltage Sense

The voltage regulator should be located as close as possible to the load. This is true especially if the output voltage sense circuitry is internal to the regulator's control device. Excessive lead length will result in an error voltage developed across the line resistance (R4').

$$V_O = V_{O(reg)} - (R_2' + R_3' + R_4')\, I_L + R_2' I_{reg}$$

$$\text{ERROR} = I_L(R_3' + R_4') - I_{reg} R_2'$$

If the voltage sense is available externally, the effect of the line resistance can be minimized. By referencing the low current external voltage sense input to the load, losses in the output line are compensated. Since the current in the sense line is very small, error introduced by its line resistance is negligible (Figure 41).

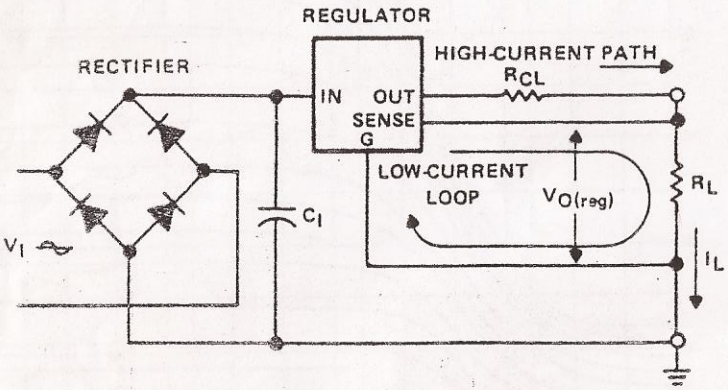

Figure 41. Proper Regulator Layout

Input Supply Design

When the power source is an ac voltage, the transformer, rectifier, and input filter design are as important as the regulator design itself for optimum system performances. This section presents input supply and filter design information for designing a basic capacitor input supply.

Transformer/Rectifier Configuration

The input supply consists of three basic sections: (1) input transformer, (2) rectifier, and (3) filter as illustrated in Figure 42. The first two sections, the transformer and the rectifier, are partially dependent upon each other because the structure of one depends upon that of the other. The most common transformer configurations and their associated rectifier circuits are illustrated in Figure 43.

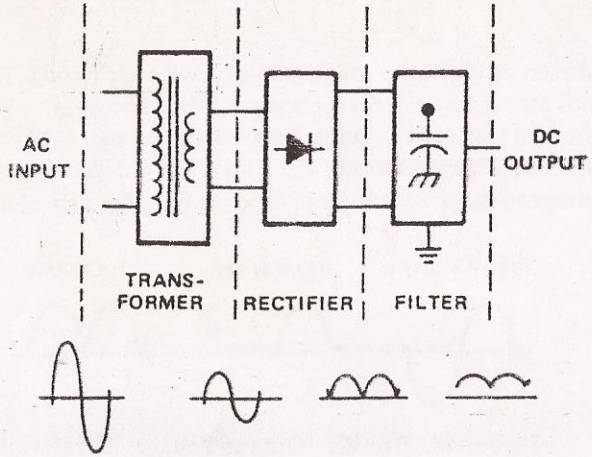

Figure 42. Input Supply

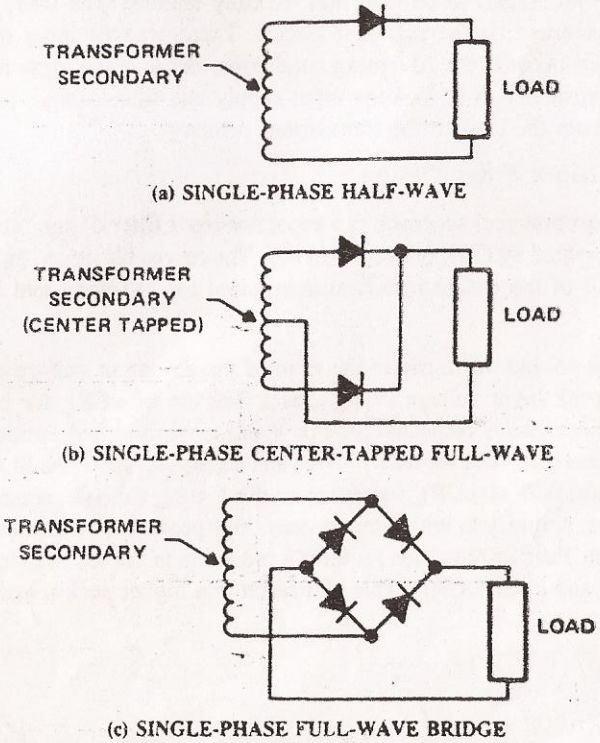

Figure 43. Input Supply Transformer/Rectifier Configurations

The particular configuration used depends upon the application. The half-wave circuit [Figure 43(a)] is used in low-current applications. This is because the single rectifier diode experiences the total load current and its conversion efficiency is less than 50%. The full-wave configurations [Figure 43(b) and 43(c)] are used for higher current application. The characteristic output voltage waveforms of these configurations are illustrated in Figure 44.

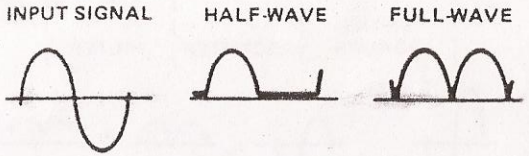

Figure 44. Rectifier Output-Voltage Waveforms

Before the input supply and its associated filter can be designed, the voltage, current, and ripple requirements of its load must be fully defined. The load, as far as the input supply is concerned, is the regulator circuit. Therefore, the input requirements of the regulator itself become the governing conditions. Because the input requirements of the regulator control circuit govern the input supply and filter design, it is easiest to work backwards from the load to the transformer primary.

Capacitor Input Filter Design

The most practical approach to a capacitor-input filter design remains the graphical approach presented by O.H. Schade[1] in 1943. The curves illustrated in Figures 45 through 48 contain all of the design information required for full-wave and half-wave rectifier circuits.

Figures 45 and 46 illustrate the ratio of the dc-output voltage developed (V_C) to the applied peak input voltage ($V_{(PK)}$), as a function of ωCR_L for half-wave and full-wave rectified signals, respectively. For a full-wave rectified application, the voltage reduction is less than 10% for $\omega CR_L > 10$ and $R_S/R_L < 0.5\%$. As illustrated, the voltage reduction decreases as ωCR_L increases or the R_S/R_L ratio decreases. Minimizing the reduction rate, contrary to initial impressions, may prove to be detrimental to the optimum circuit design. Further reduction requires a reduction in the series to load resistance ratio (R_S/R_L) for any given ωCR_L. This will result in a higher peak-to-average current ratio

[1] O.H. Schade, "Analysis of Rectifier Operation", *Proc. IRE.*, VOL. 31, 343, 1943.

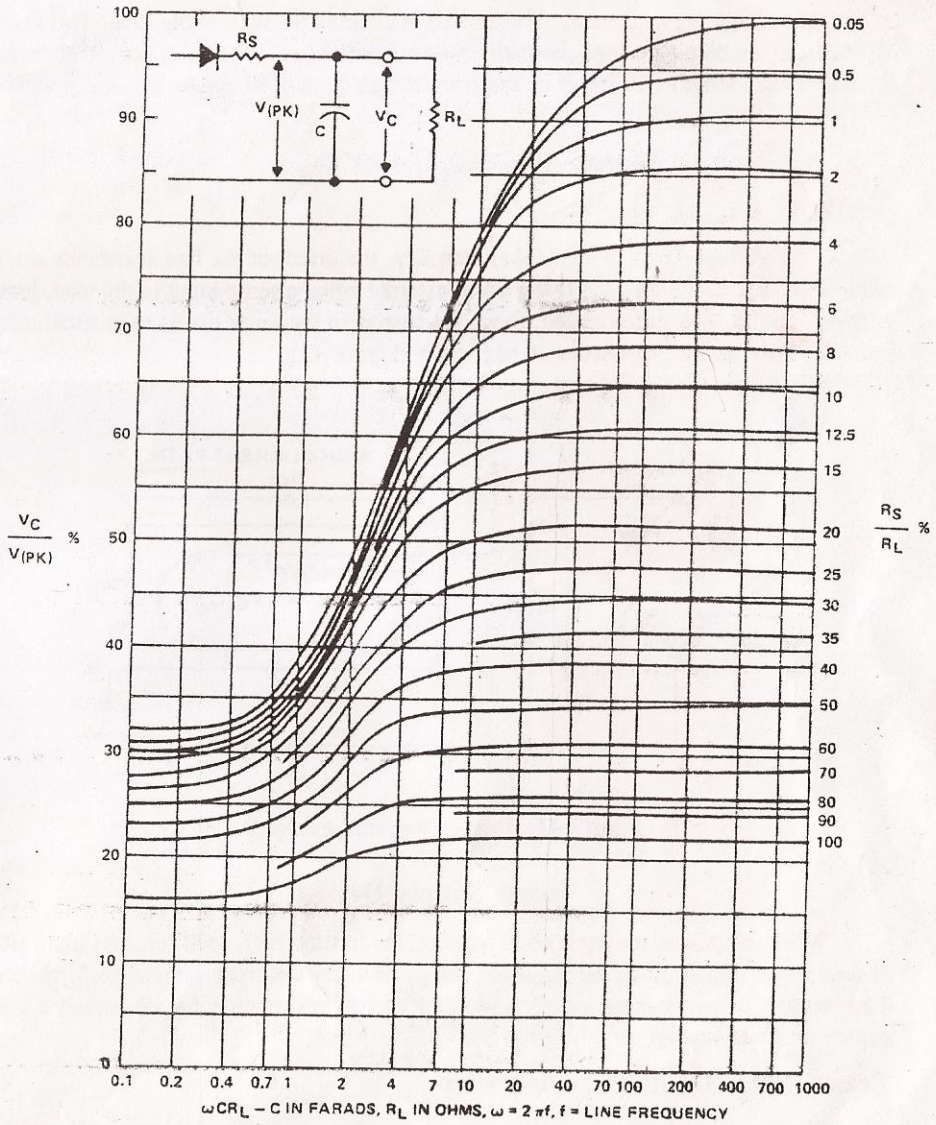

Figure 45. Relation of Applied Alternating Peak Voltage to Direct Output Voltage in Half-Wave Capacitor-Input Circuits (From O.H. Schade, *Proc. IRE*, Vol. 31, p. 343, 1943)

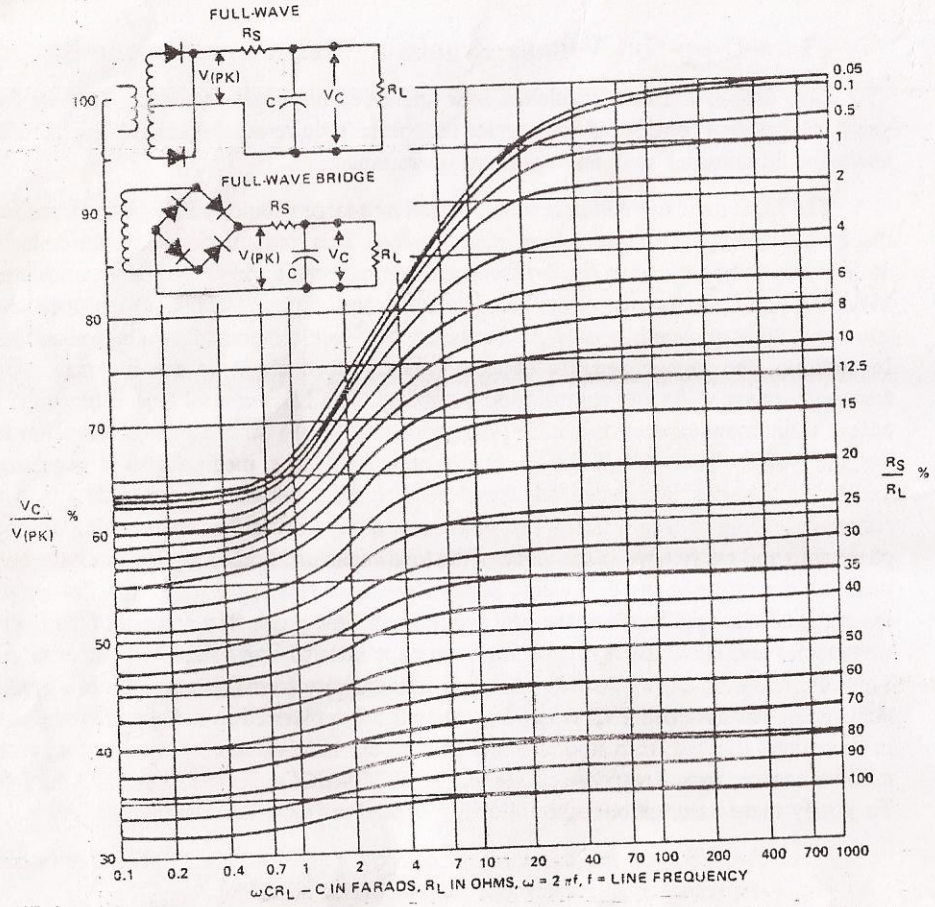

Figure 46. Relation of Applied Alternating Peak Voltage to Direct Output Voltage in Full-Wave Capacitor-Input Circuits
(From O.H. Schade, *Proc. IRE*, Vol. 31, p. 344, 1943)

through the rectifier diodes (see Figure 47). In addition, and probably of more concern, this increases the surge current experienced by the rectifier diodes during turn-on of the supply. It is important to realize that the surge current is limited only by the series resistance R_S.

$$I_{SURGE} = \frac{V_{SEC(PK)}}{R_S}$$

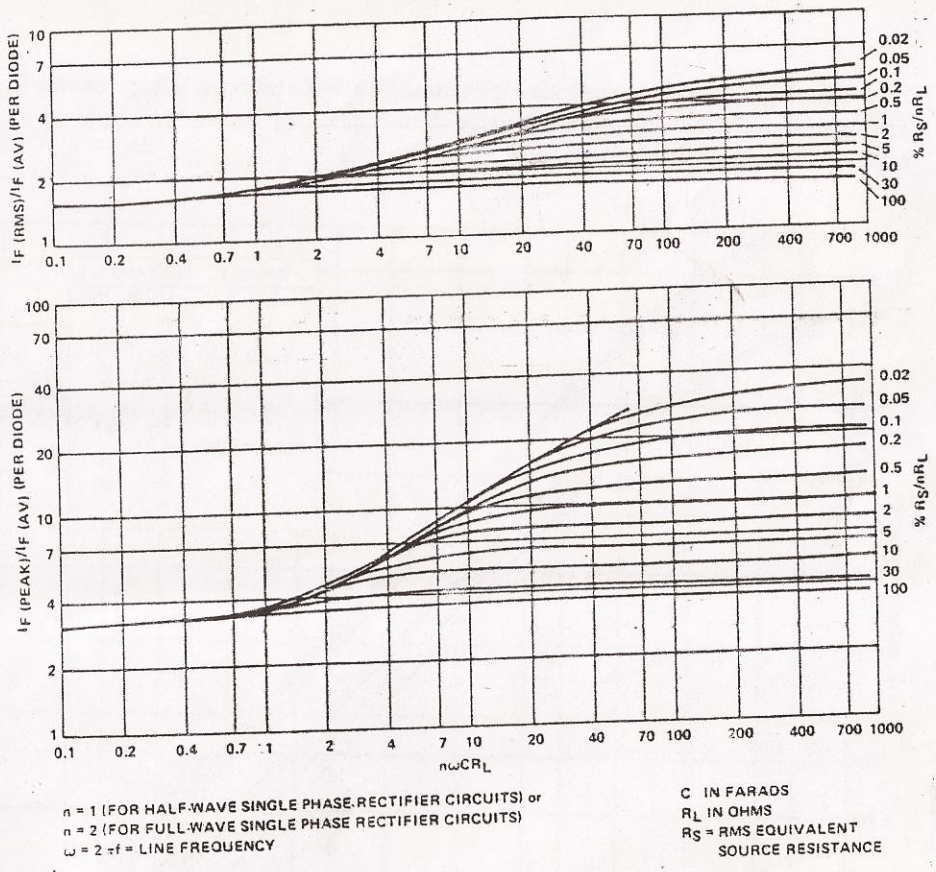

Figure 47. Relation of RMS and Peak-to-Average Diode Current in Capacitor-Input Circuits
(From O.H. Schade, *Proc. IRE*, Vol. 31, p. 345, 1943)

In order to control the surge current, additional resistance is often required in series with each rectifier. It is evident that a compromise must be made between the voltage reduction and the rectifier current ratings. The maximum instantaneous surge current is $V_{(pk)}/R_S$. The time constant (τ) of capacitor C is $\tau \cong R_S C$. As a rule of thumb, the surge current will not damage the diode if: $I_{SURGE} < I_{F(SURGE)max}$ and $\tau < 8.3$ ms

Figure 48 illustrates the relationships between the ripple factor r_f, ωCR_L, and R_S/R_L. The ripple factor is the ratio of the rms value of the ripple component of the output voltage, expressed as a percent of the nominal dc output voltage.

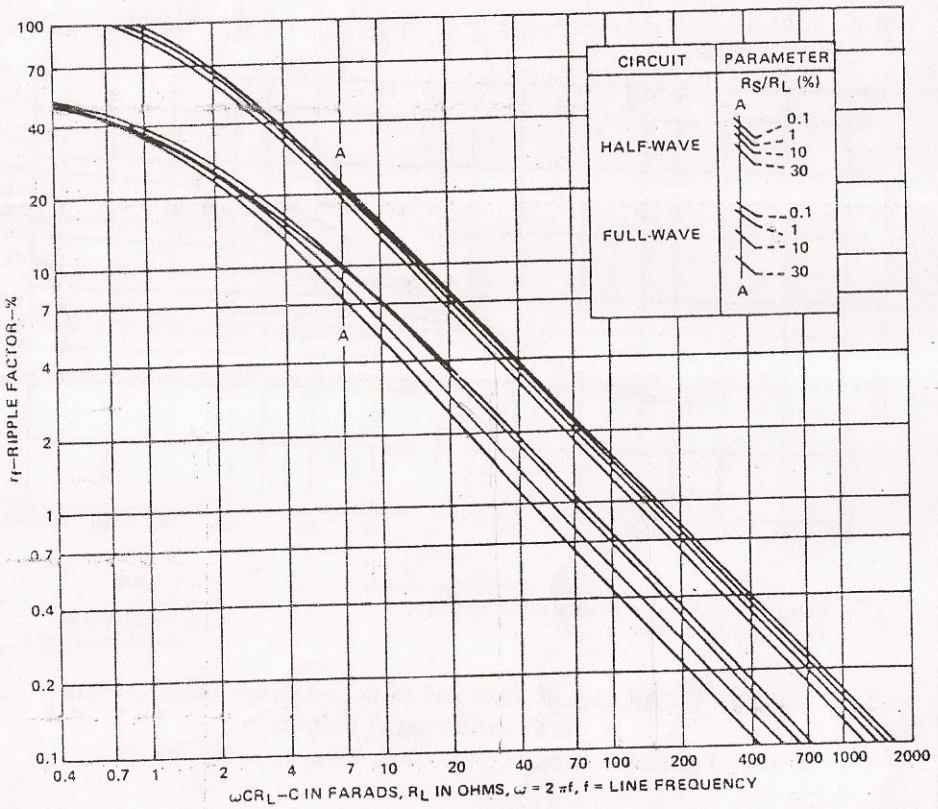

Figure 48. Root-Mean-Square Ripple Voltage for Capacitor-Input Circuits (From O.H. Schade, *Proc. IRE*, Vol. 31, p. 346, 1943)

Low Drop-Out Voltage Regulator Design Considerations

Low drop-out (LDO) regulators have increased in popularity during the past five years. This design consideration discusses the special requirements that LDO applications users should consider to obtain optimum operations.

The LDO regulator design is characterized by a large geometry PNP output transistor that is used to conduct the power from input to output. This configuration places the collector at the output while reducing the overhead voltage required to drive the output transistor. Since the output impedance of the collector is several thousand ohms, a high open loop gain must be maintained to reach the necessary MΩ requirements of a voltage regulator. In addition, the high-impedance collector reactive-load elements can introduce low-frequency-phase shifts and control-loop instability. The LDO control loop is designed to accept a minimum capacitive loading (approximately 10 μF) for complete stablization for the low-power LDO (TL750LXX series) regulators. The medium-power regulators TL750MXX series introduced additional requirements because of the wide range of restistive loading (5 Ω to infinity) that have resulted in a broad frequency swing in loop-phase shift and restrictions in the value of the load capacitor. Some assumptions have been made on the range of the Equivalent Series Resistance (ESR) the load capacitor should exhibit in certain applications. The ESR would cancel the phase shift at some of the higher frequencies and allow the regulator design to have internal fixed-stability compensation. This compensation allows the load capacitor value to vary from the minimum to a greater than value. The assumed ESR is approximately 0.5 Ω (polarized solid tantalum to be used in this range) derived from studies of the ESR of the standard available capacitors at the minimum value. Actual response curves for the TL750MXX series are shown in Figure 49. To justify these assumptions, the following conditions must be stated:

1. The ESR of the capacitor is directly related to the application (wiring resistance of regulator and load increases the ESR).
2. The loop will become unstable if the ESR is forced too low.
3. The stability will improve with an increase in load-capacitor value.

Caution must be exercised in some medium-power LDO applications that require stable regulation over a large load range because mV-level oscillations may be exhibited at the higher load-resistance ranges. To eliminate the oscillations, a larger capacitor value (approximately 100-μF maximum) should be selected or the ESR value should be increased (to approximately 2 Ω). The circuit bandwidth is decreased by the larger value capacitor and the loop bandwidth is increased with the higher ESR. This provides a faster transient load response and offsets some of the loss from the capacitor ESR.